LECTURE 14
Numerical Dissipation and Dispersion

Reference: J. C. Tannehill, D. A. Anderson & R. H. Pletcher, Computational Fluid Mechanics and Heat Transfer, 2nd ed., Taylor & Francis, 1997. Chapter 4, Sections 4.1.1 to 4.1.8.

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The Modified Equation

A difference scheme does not solve the equation it was built from.

It solves that equation plus its own truncation error, and the **modified** equation we obtain is the result written out as a differential equation.

We need to follow 3 steps:

- Expand every term of the difference equation about (i, n).
- Subtract the original equation, leaving the truncation error.
- Remove every time derivative from the error using the equation itself.

$$\frac{\partial u}{\partial t} + c\,\frac{\partial u}{\partial x} = \nu_2\,\frac{\partial^2 u}{\partial x^2} + \nu_3\,\frac{\partial^3 u}{\partial x^3} + \nu_4\,\frac{\partial^4 u}{\partial x^4} + \dots$$

The coefficients depend on c, Δx and Δt.

The ν terms control what the computed profile will actually show.

Modified Equation of the Upwind Method

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} + c\,\frac{u_i^n - u_{i-1}^n}{\Delta x} = 0$$

Step 1: we expand the two differences.

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = u_t + \frac{\Delta t}{2} u_{tt} + \ldots, \qquad \frac{u_i^n - u_{i-1}^n}{\Delta x} = u_x - \frac{\Delta x}{2} u_{xx} + \ldots$$

Step 2: assemble and subtract the original equation.

$$u_t + c\,u_x = -\frac{\Delta t}{2} u_{tt} + \frac{c\,\Delta x}{2} u_{xx} + \ldots$$

Step 3: remove the time derivative with $u_{tt} = c^2 u_{xx}$, which follows from the equation itself.

$$u_t + c\,u_x = \underbrace{\frac{c\,\Delta x}{2}(1 - C)}_{\nu_2}\, u_{xx} + \underbrace{\frac{c\,\Delta x^2}{6}(1 - C)(2C - 1)}_{\nu_3}\, u_{xxx} + \ldots$$

Dispersion Relations of the Model Equations

We substitute the same Fourier mode:

$$u = \hat{u}\, e^{\,\mathrm{i}(kx-\omega t)}, \qquad \frac{\partial^m}{\partial x^m} \to \mathrm{i}^{\,m} k^{\,m}$$

$$\Longrightarrow \qquad u = \hat{u}\; e^{\,\mathrm{i}k(x-(\Re\omega/k)\,t)}\;\; e^{\,\Im\omega\, t}$$

so that every derivative becomes an algebraic factor:

$$u = e^{\,i(kx-\omega t)} \qquad \Longrightarrow \qquad \frac{\partial}{\partial t} \to -i\omega, \quad \frac{\partial}{\partial x} \to ik, \quad \frac{\partial^2}{\partial x^2} \to -k^2, \quad \frac{\partial^3}{\partial x^3} \to -ik^3$$

Even derivative: $u_t + c\,u_x = \nu_2\, u_{xx}$

$$-i\omega + ick = -\nu_2 k^2 \quad \Longrightarrow \quad \omega = ck - i\,\nu_2 k^2 \quad \Longrightarrow \quad u = e^{-\nu_2 k^2 t}\, e^{\,ik(x-ct)}$$

Odd derivative: $u_t + c\,u_x = \nu_3\, u_{xxx}$

$$-i\omega + ick = -i\,\nu_3 k^3 \quad \Longrightarrow \quad \omega = ck + \nu_3 k^3 \quad \Longrightarrow \quad u = e^{\,ik\left[\,x-(c+\nu_3 k^2)\,t\,\right]}$$

Dissipation and Dispersion

The imaginary part of ω sets the amplitude, the real part sets the speed:

$$\nu_2\, u_{xx} \quad \Rightarrow \quad \omega = ck - \mathrm{i}\nu_2 k^2, \quad \text{amplitude} \times e^{-\nu_2 k^2 t}, \quad c_{\text{num}} = c$$

$$\nu_3\, u_{xxx} \Rightarrow \quad \omega = ck + \nu_3 k^3, \quad \text{amplitude fixed}, \quad c_{\text{num}} = c + \nu_3 k^2$$

With ν_2 the frequency is complex, so every mode decays at the rate $\nu_2 k^2$ and all of them still travel at c.

With ν_3 it stays real, so no mode decays and each one travels at $c + \nu_3 k^2$. These are the two panels of the previous slide.

i to an even power is real and to an odd power is imaginary

Thus, **even derivatives collect into the amplitude and the odd ones into the speed.**

Dissipation and Dispersion on a Step

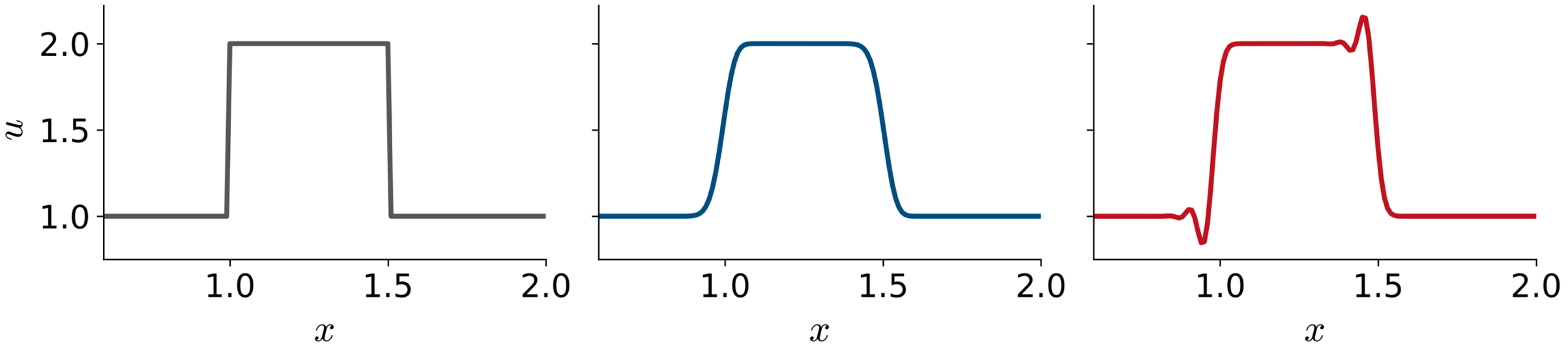
Here we compute the advection using the **Upwind** and **Lax-Wendroff** methods.

201 points, $C = 0.8$, $t = 0.5$. Compare Tannehill, Figure 4.1.

Exact advection

Dissipation errors (1st-order methods)

Dispersion errors (2nd-order methods)

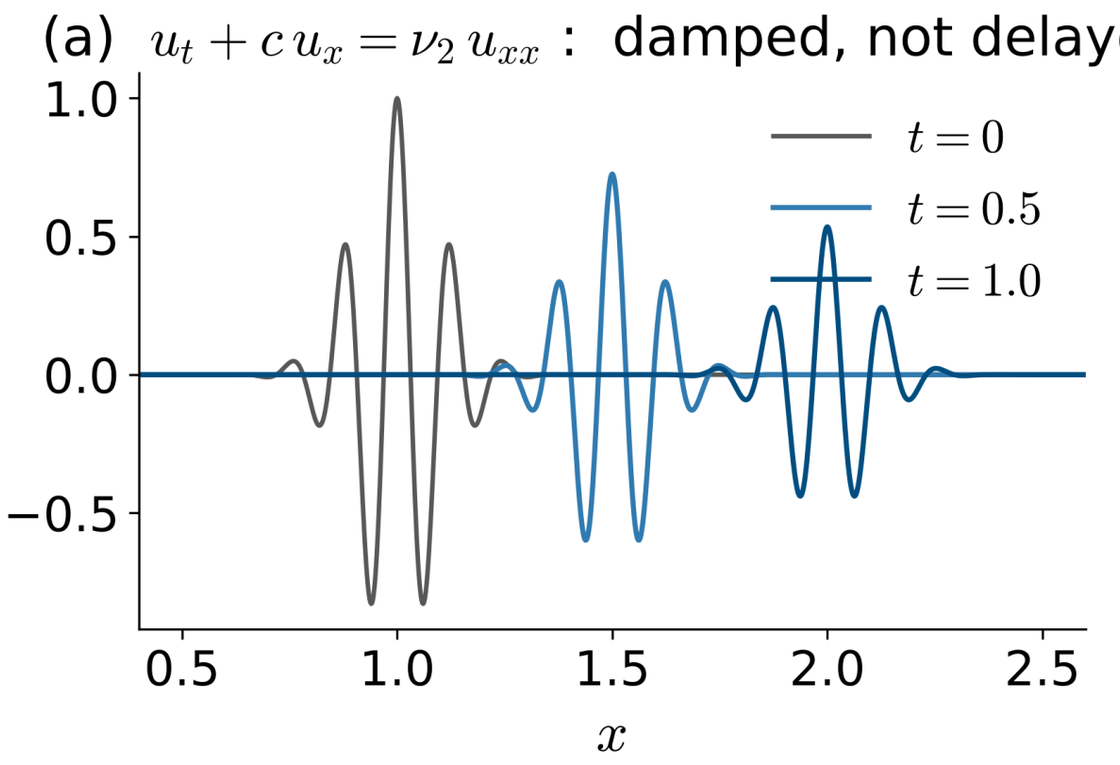


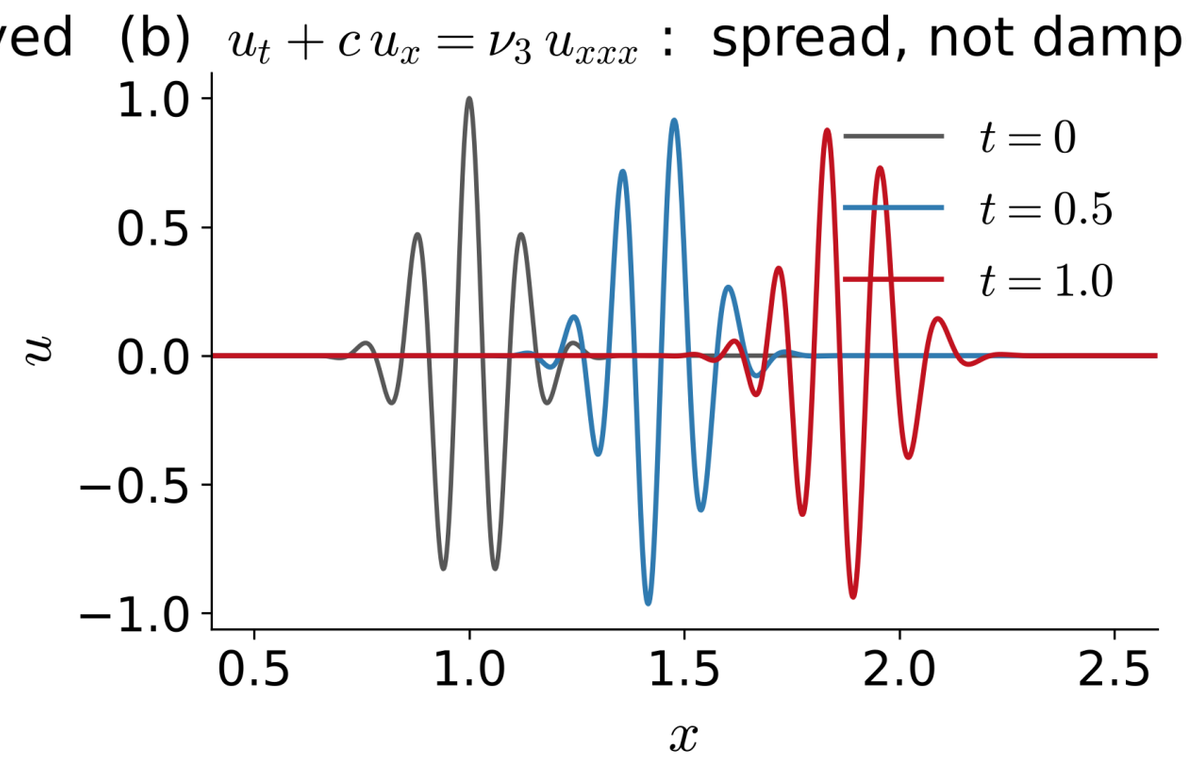
Dissipation: the even-order terms of the modified equation, $\nu_2\, u_{\mathrm{xx}}$ and $\nu_4\, u_{\mathrm{xxxx}}$. They push $|G|$ below 1, so every mode loses amplitude and the step is smeared. The speed is untouched.

Dispersion: the odd-order terms, $\nu_3\, u_{\mathrm{xxx}}$ and $\nu_5\, u_{\mathrm{xxxxx}}$. They leave $|G|$ alone and shift the phase, so each mode travels at its own speed and the short waves fall behind.

Example: Effect of the Error Terms on a Wave Packet

A Gaussian packet advanced by the exact solution of each model equation, evaluated with an FFT on 2048 points.





Dissipation, panel (a): $u_t + c\,u_x = \nu_2\,u_{xx}$, an advection-diffusion equation. Every mode decays at the rate $\nu_2 k^2$, and all of them still travel at c.

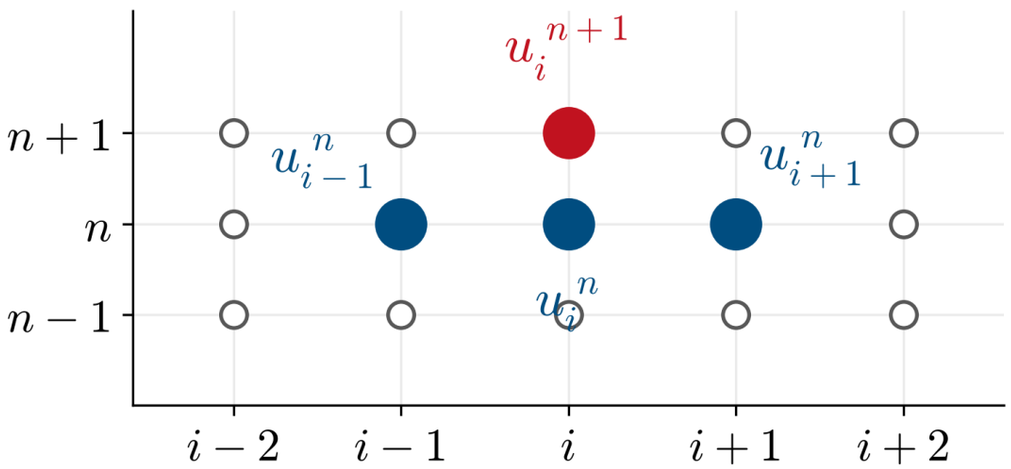
Dispersion, panel (b): $u_t + c\,u_x = \nu_3\,u_{xxx}$, the linearised Korteweg-de Vries equation. No mode decays, but each travels at $c + \nu_3 k^2$, so the packet spreads.

Euler Explicit Methods: FTCS method

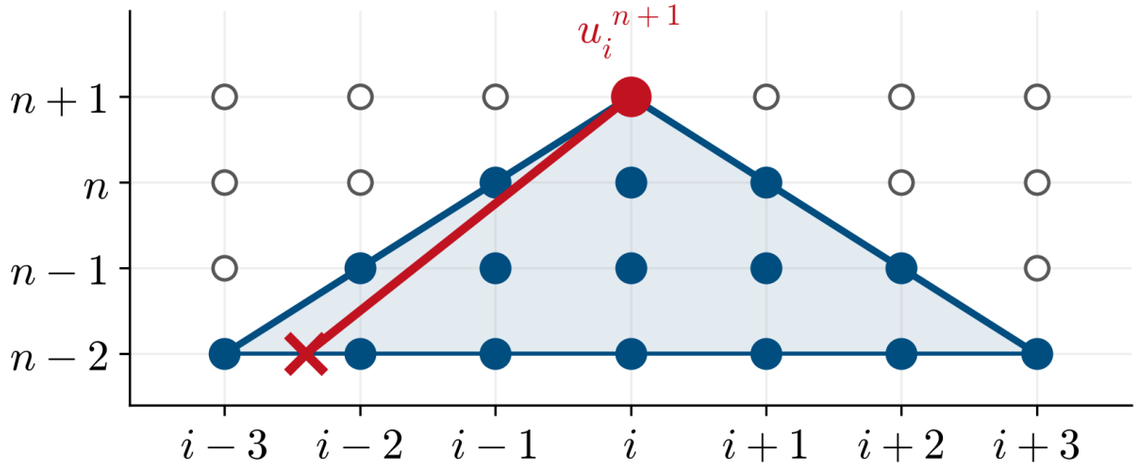
For mathematical reasons, we start with the forward in time, centred in space (FTCS):

$$u_i^{n+1} = u_i^n - \frac{C}{2}\left(u_{i+1}^n - u_{i-1}^n\right), \qquad |G|^2 = 1 + C^2 \sin^2\theta \; > \; 1$$

stencil: centred in space



numerical domain of dependence (blue) and the characteristic (red), $C = 0.8$



As mentioned in lecture 12, the FTCS method is **unconditionally unstable.** Here, we see that this is because **t**he modified equation has $\nu_2 = -c^2\Delta t/2$, which is a backwards diffusion that amplifies every mode.

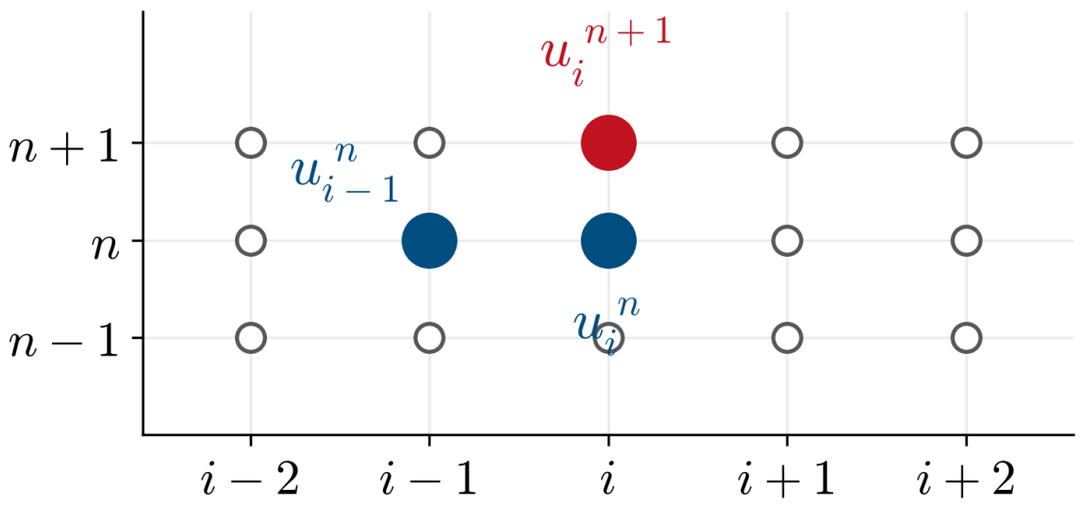
Upwind Method (Upstream, FTBS method)

We go backward in space: the stencil reaches towards the side the wave arrives from, and ν_2 becomes positive for $0 < C < 1$.

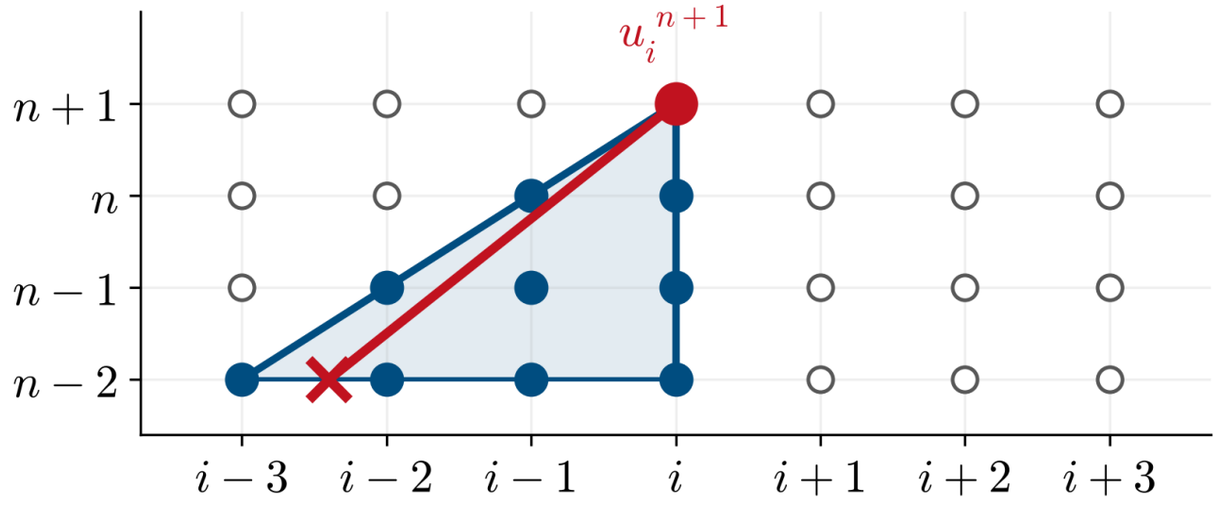
$$u_i^{n+1} = u_i^n - C\left(u_i^n - u_{i-1}^n\right)$$

$$|G|^2 = 1 - 2C(1-C)(1-\cos\theta), \qquad \nu_2 = \frac{c\,\Delta x}{2}(1-C)$$

Stencil



Domain of dependence
Characteristic line



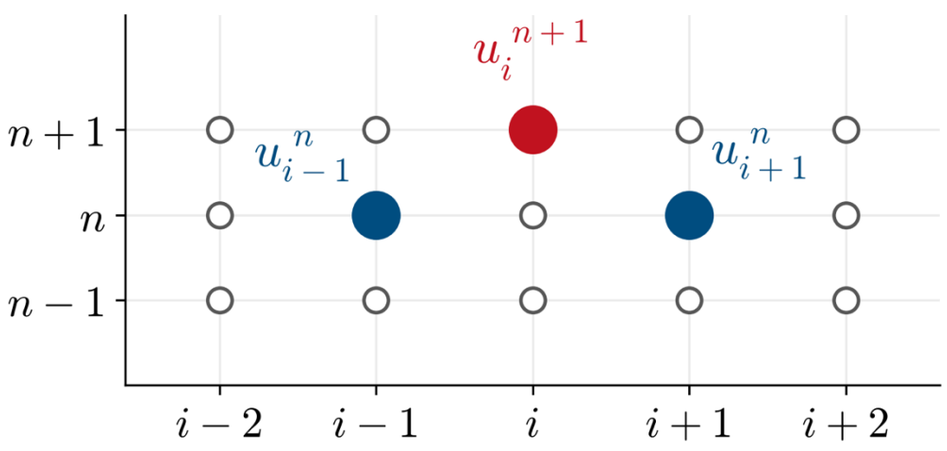
Note that the domain of dependence is one-sided. It matches the direction the information travels, so it captures the motion.

Lax Method

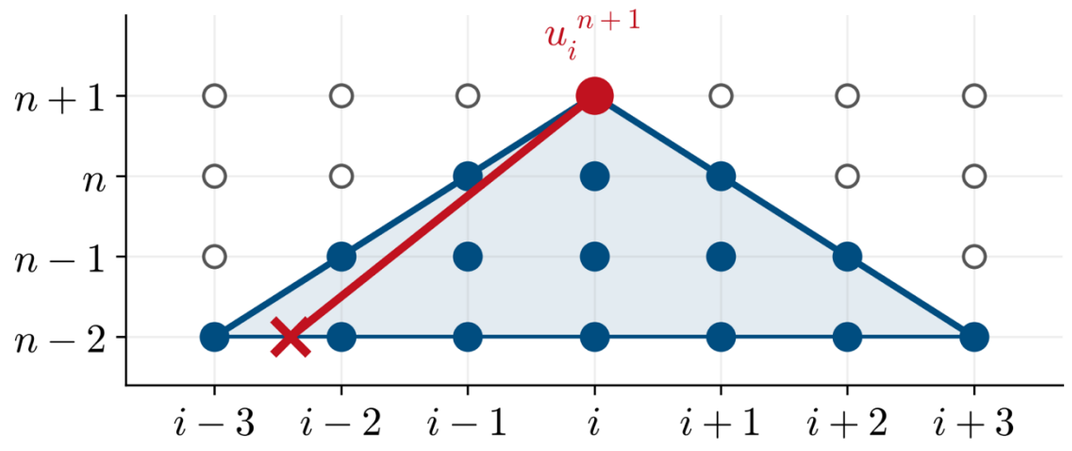
The construction starts with the FTCS method, but the old value in the time derivative is replaced by the average of its neighbours.

$$u_i^{n+1} = \frac{1}{2}\left(u_{i+1}^n + u_{i-1}^n\right) - \frac{C}{2}\left(u_{i+1}^n - u_{i-1}^n\right)$$

Stencil



Domain of dependence
Characteristic line



$$G = \cos\theta - \mathrm{i}\,C\sin\theta, \qquad |G|^2 = 1 - (1 - C^2)\sin^2\theta, \qquad \nu_2 = \frac{\Delta x^2}{2\,\Delta t}(1 - C^2)$$

The truncation error is $O[\Delta t, (\Delta x)^2/\Delta t]$:

Reducing Δt at fixed mesh makes the method more diffusive, not less.

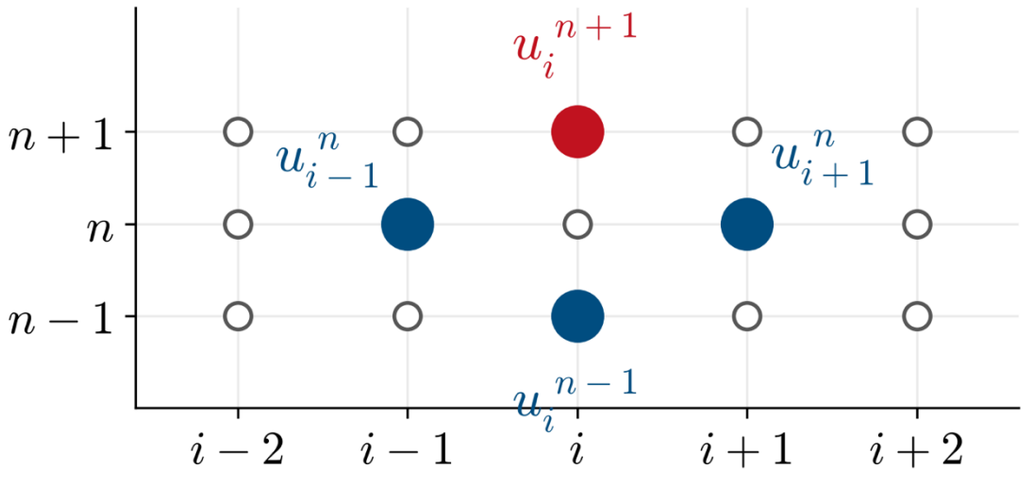
Leap Frog Method

The construction is centred in time as well as in space, over 3-time levels.

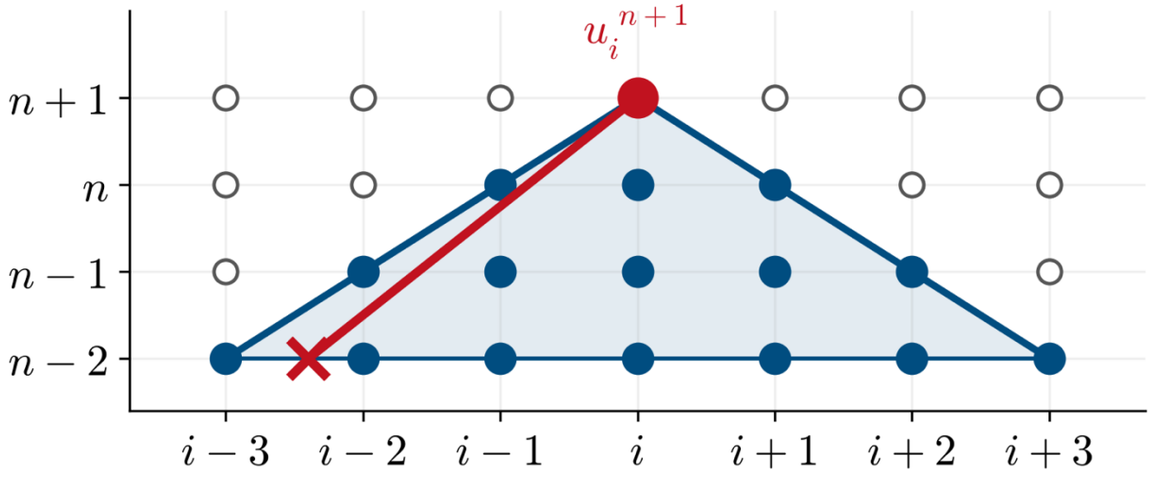
$$u_i^{n+1} = u_i^{n-1} - C\left(u_{i+1}^{n} - u_{i-1}^{n}\right)$$

$$G = \pm\left(1 - C^2 \sin^2\theta\right)^{1/2} - \mathrm{i}\, C \sin\theta$$

Stencil



Domain of dependence
Characteristic line



$$|G| = 1 \ \text{ for all } \theta, \quad C \le 1, \qquad u_t + c\,u_x = \frac{c\,\Delta x^2}{6}\left(C^2 - 1\right) u_{xxx} + \ldots$$

No even derivative appears:

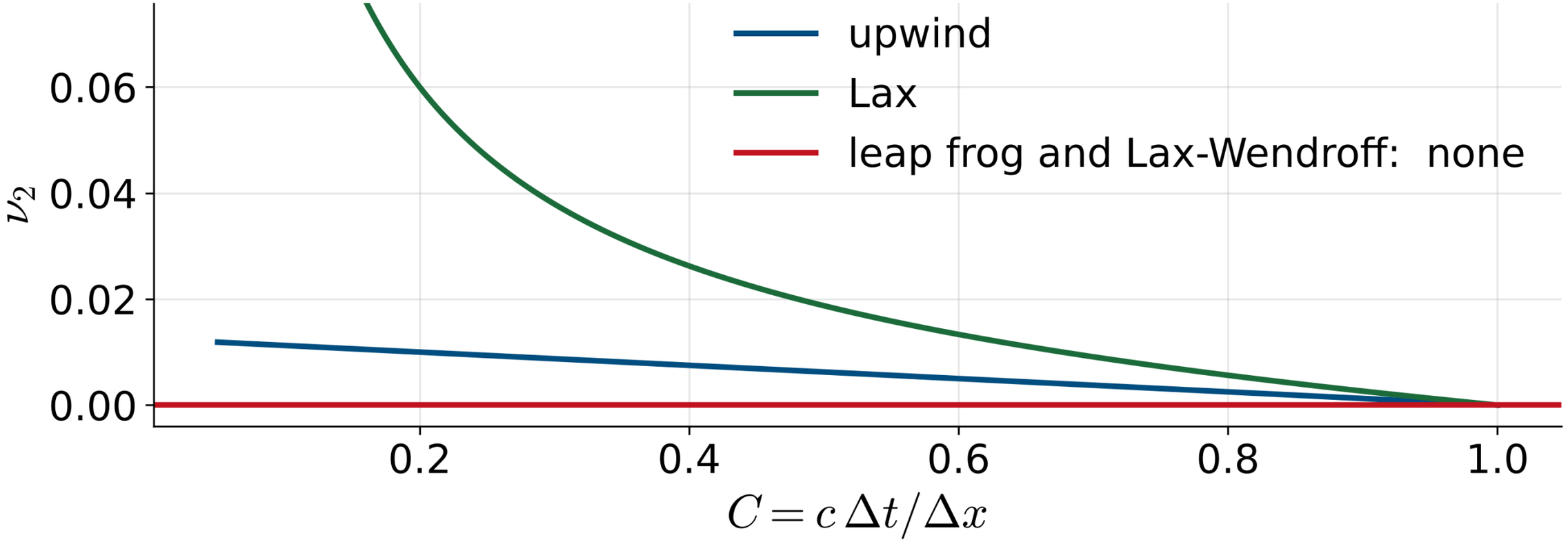
The method is neutrally stable and purely dispersive.

It needs a starting procedure and a filter for its two solutions.

Coefficient of the Second-Derivative Term

$$u_t + c\,u_x = \underbrace{\frac{c\,\Delta x}{2}(1-C)}_{\nu_2}\,u_{xx} + \underbrace{\frac{c\,\Delta x^2}{6}(1-C)(2C-1)}_{\nu_3}\,u_{xxx} + \dots$$

To assess diffusion, we can plot the coefficient that accompanies u_{xx} in the **modified equation** at a fixed Δx.



- The upwind method improves as C approaches 1.
- The Lax method is worst at small C.

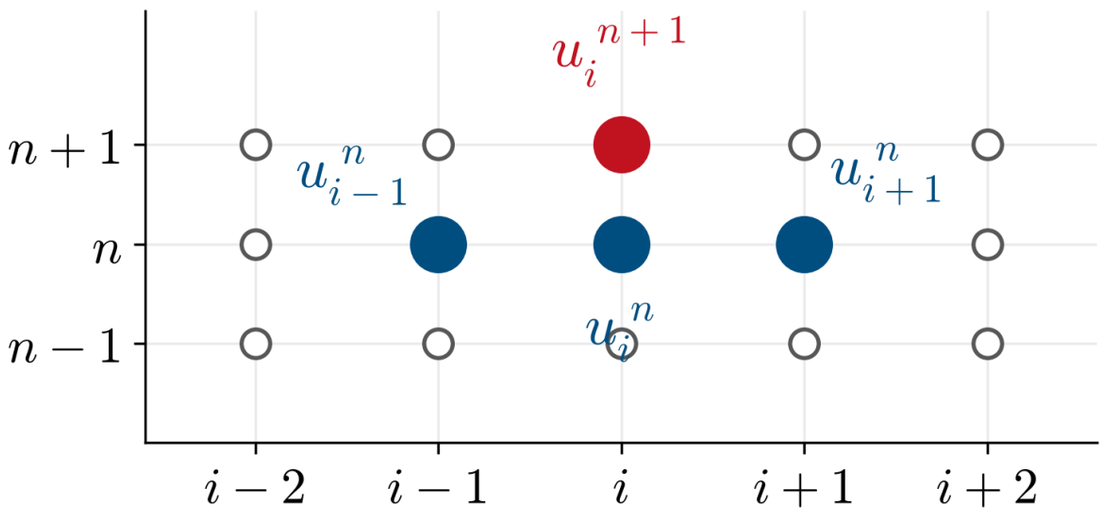
Lax-Wendroff Method

For its construction, we expand in time, replace both time derivatives from the equation, and only then difference the space derivatives.

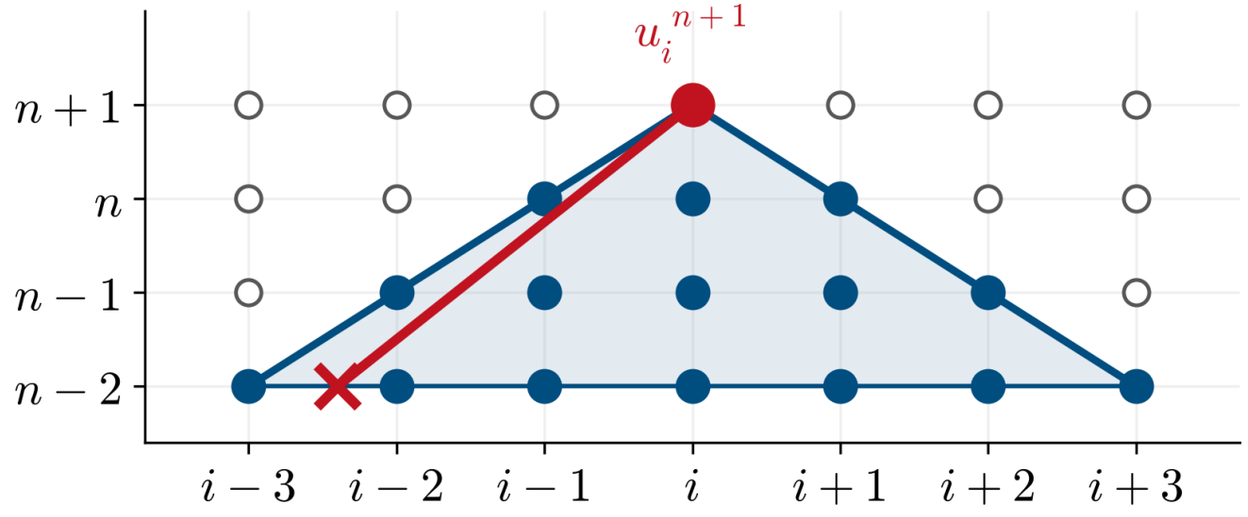
$$u_i^{n+1} = u_i^n + \Delta t\, u_t + \frac{\Delta t^2}{2} u_{tt} + \ldots, \qquad u_t = -c\, u_x, \qquad u_{tt} = c^2 u_{xx}$$

$$\boxed{u_i^{n+1} = u_i^n - \frac{C}{2}\left(u_{i+1}^n - u_{i-1}^n\right) + \frac{C^2}{2}\left(u_{i+1}^n - 2u_i^n + u_{i-1}^n\right)}$$

Stencil



Domain of dependence
Characteristic line



You can see the full construction here: https://github.com/wbandabarragan/computational-physics-2/blob/main/unit-5/507-wave-equation-Lax-Wendroff-scheme.ipynb

Lax-Wendroff: Amplification Factor and Modified Equation

$$G = 1 - C^2\,(1 - \cos\theta) - \mathrm{i}\,C \sin\theta$$

$$|G|^2 = \left[1 - C^2(1 - \cos\theta)\right]^2 + C^2 \sin^2\theta$$

$$= 1 - 4\,C^2\left(1 - C^2\right)\sin^4(\theta/2)$$

Stable for C ≤ 1: the damping goes as the fourth power of $\sin(\theta/2)$, so long waves are almost untouched.

$$u_t + c\,u_x = -\frac{c\,\Delta x^2}{6}\left(1 - C^2\right) u_{xxx} + \ldots$$

$$c_{\mathrm{num}} = c\left[1 - \frac{1 - C^2}{6}\,(k\,\Delta x)^2\right] \;<\; c$$

The same leading term as leap frog:

Short waves run slow, so the oscillations sit behind the discontinuity, never ahead of it.

Two-Step Lax-Wendroff and MacCormack Methods

These are modifications of the previous method, which for a system would need the flux Jacobian. Both reduce to the 1-step formula here, so all three share G.
Neither method writes the second derivative down:

Two-step Lax-Wendroff:

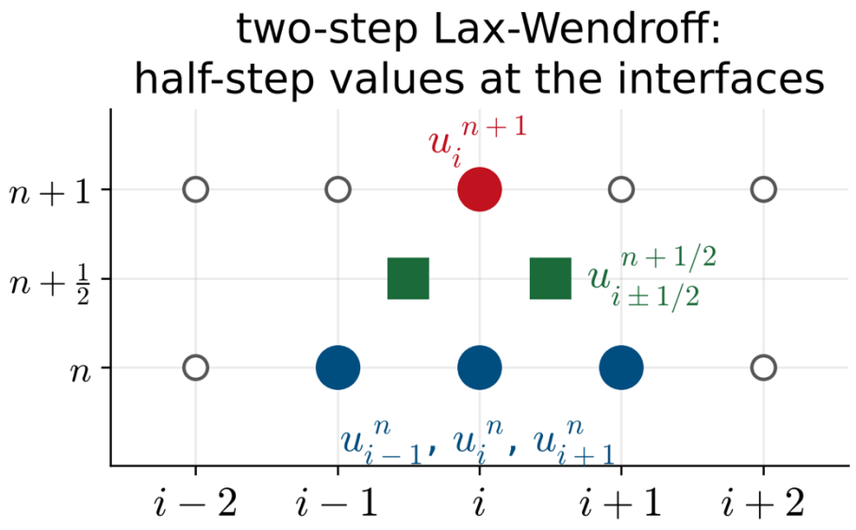
$$u_{i+1/2}^{n+1/2} = \frac{1}{2}\left(u_{i+1}^{n} + u_{i}^{n}\right) - \frac{C}{2}\left(u_{i+1}^{n} - u_{i}^{n}\right)$$

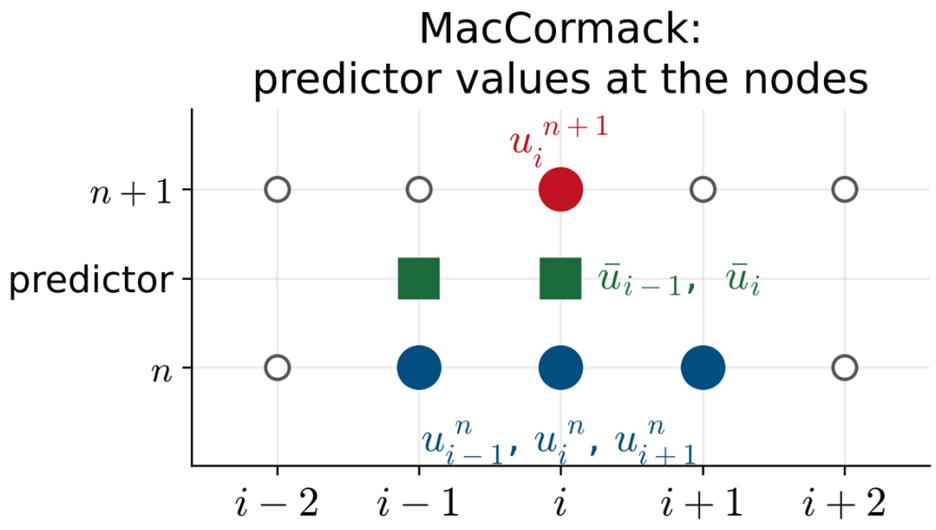
$$u_{i}^{n+1} = u_{i}^{n} - C\left(u_{i+1/2}^{n+1/2} - u_{i-1/2}^{n+1/2}\right)$$

MacCormack:

$$\bar{u}_i = u_i^n - C\left(u_{i+1}^n - u_i^n\right)$$

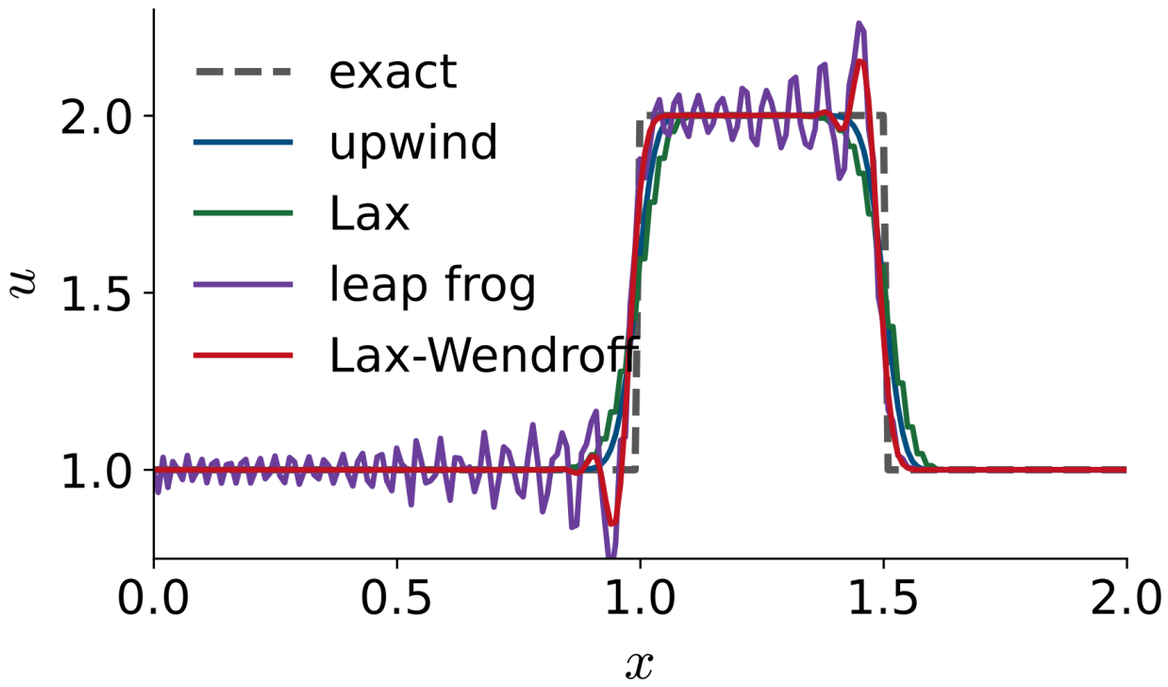
$$u_i^{n+1} = \frac{1}{2}\left[u_i^n + \bar{u}_i - C\left(\bar{u}_i - \bar{u}_{i-1}\right)\right]$$

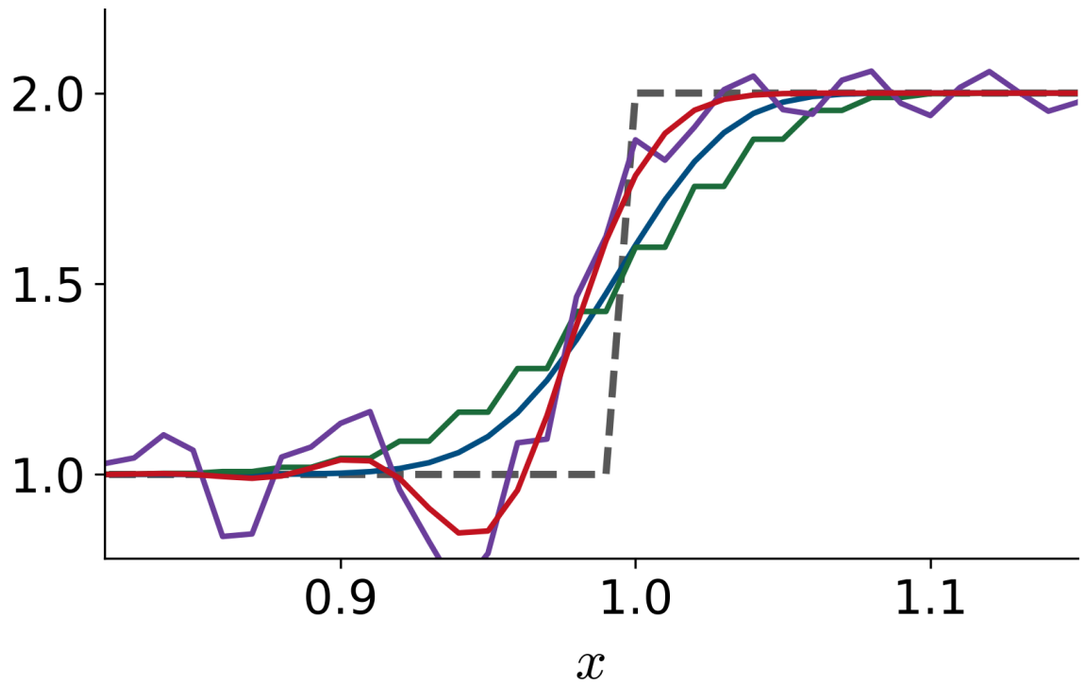




Solutions for a top-hat advection problem (step):

The solution is computed here on 201 points at $C = 0.8$, marched to $t = 0.5$ from the square wave of nb08.



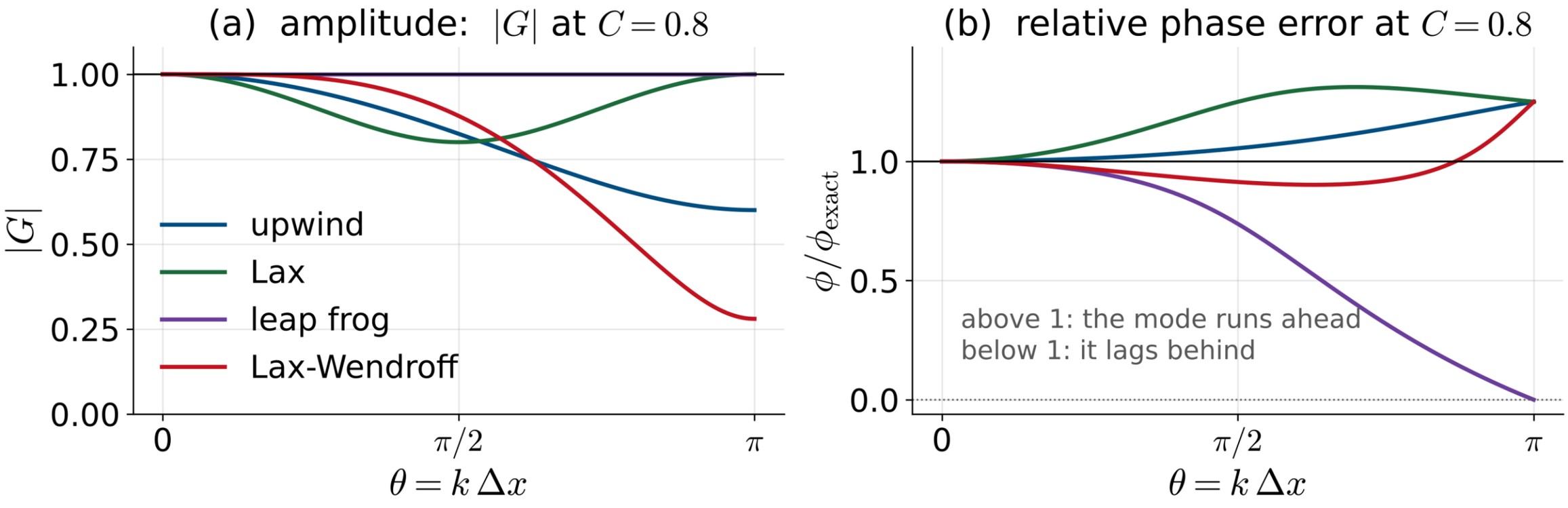


Zooming into the step shows the amplitudes under each scheme.

See the integration comparisons here: https://github.com/wbandabarragan/computational-fluid-dynamics/blob/main/notebooks/nb08_dissipation_dispersion.ipynb

Amplitude and Phase Errors

We can also look into the amplitude and phase errors, which are evaluated from the amplification factors at $C = 0.8$.



The left plot shows dissipation errors.

The right plot shows dispersion errors.

Note that the leap frog method keeps $|G| = 1$ at every wavenumber.

Comparison of Methods

Method	Truncation error	Leading error term	Error type	Used today
Euler explicit (FTCS)	$O[\Delta t, (\Delta x)^2]$	$\nu_2 < 0$	unstable for every C	no
Upwind (first-order upstream, FTBS)	$O[\Delta t, \Delta x]$	$\frac{1}{2}\, c\, \Delta x\, (1 - C)\, u_{xx}$	dissipative	yes, the first-order building block of Godunov-type schemes
Lax	$O[\Delta t, (\Delta x)^2/\Delta t]$	$(\Delta x^2/2\Delta t)(1 - C^2)\, u_{xx}$	dissipative	as the local Lax, or Rusanov, flux in finite-volume codes
Leap frog	$O[(\Delta t)^2, (\Delta x)^2]$	$-(c\, \Delta x^2/6)(1 - C^2)\, u_{xxx}$	dispersive only, $\lvert G\rvert = 1$	yes, in atmosphere and ocean models, with a time filter
Lax-Wendroff 1-step and 2-step	$O[(\Delta t)^2, (\Delta x)^2]$	$-(c\, \Delta x^2/6)(1 - C^2)\, u_{xxx}$	dispersive	as the high-order flux inside limited, TVD schemes
MacCormack	$O[(\Delta t)^2, (\Delta x)^2]$	as Lax-Wendroff	dispersive	in legacy aerodynamic codes; superseded by Godunov-type methods

Summary

- The so-called modified equation is what a scheme actually solves: the original equation plus the truncation error, with the time derivatives removed.

- Even-order error terms change the amplitude and leave the speed alone.
- Odd-order terms do the reverse.

First-order methods lead with $u_{\{xx\}}$ and smear a step.
The Lax method is the more diffusive of the two, and gets worse as C falls.

- Leap-frog and Lax-Wendroff are second order and lead with $u_{\{xxx\}}$, so they keep a step sharp and oscillate behind it. Leap frog has no dissipation at all.

Methods in use today:
- First-order upwind and the local Lax (Rusanov) flux are building blocks
- Leap frog is used in geophysical codes
- Lax-Wendroff as the high-order flux inside limited schemes.
- MacCormack survives mainly in legacy codes.